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Clinical NLP for Morbidity Extraction and Suggested

NHS ICD-10 coding

Comparing Transformer-Centred Bio_ClinicalBERT and

Terminology-Centred Apache cTAKES Pipelines

on MIMIC-IV Discharge Summaries

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Dissertation submitted in partial fulfilment for the degree of
Master of Science in Big Data

September 2026

Abstract

Problem: Electronic medical records contain rich information about patients' diseases and medical histories, but much of it remains buried in unstructured clinical text. Automatically converting this information into structured morbidity data and linking it to standardised NHS ICD-10 codes could make patient histories easier to retrieve, interpret and reuse.

This study compares two contrasting approaches to this task: a transformer-centred Bio_ClinicalBERT pipeline within EHRKit and a terminology-centred Apache cTAKES pipeline.

Methodology: Bio_ClinicalBERT was developed using 100 manually annotated, patient-disjoint MIMIC-IV discharge summaries, while a further 60 records supported error-driven refinement of both pipelines. Once development was complete, both systems were frozen. A preliminary evaluation on 20 previously unseen records was then followed by a substantially larger evaluation on 200 unseen discharge summaries containing **2,103 Gold-standard morbidity concepts**. Extracted morbidities were normalised using UMLS and SNOMED CT UK resources and linked to NHS ICD-10 5th Edition families.

Results: The preliminary evaluation revealed contrasting system behaviour, and the 200-record experiment showed that this pattern remained stable at a larger scale. Bio_ClinicalBERT retained very high precision (**90.07%**) but lower recall (**65.53%**), with an F1 score of **75.86%**. cTAKES achieved lower precision (**71.62%**) but higher recall (**75.61%**), with an F1 score of **73.56%**.

For ICD-10 assignment, cTAKES also achieved higher coding accuracy (**92.47%** vs **88.91%**) and correct-code coverage (**87.41%** vs **83.28%**). Despite their different error profiles, **87.59%** of all Gold-standard morbidities were detected by at least one of the two systems.

Conclusion: The larger evaluation reinforced the distinction between the two approaches: Bio_ClinicalBERT was consistently more selective and precise, whereas cTAKES captured a broader range of morbidities and performed better in terminology-based coding. *Their complementary behaviour suggests that combining transformer-based contextual recognition with terminology-driven extraction and normalisation may provide a more complete and reliable solution than either pipeline alone.*

Attestation

I understand the nature of plagiarism, and I am aware of the University's academic integrity policy.

I certify that this dissertation reports work completed by me during the MSc dissertation, except where external data, software, pre-trained models and terminology resources are explicitly acknowledged and cited.

The project used the MIMIC-IV-Note dataset [1], Apache cTAKES [5], Bio_ClinicalBERT [6], EHR-Kit [7], medSpaCy and ConText [16], [17], UMLS [4], SNOMED CT UK and NHS ICD-10 mapping resources [2], [3], [15]. These resources were not created by me and were used under their applicable access conditions and licences.

My contribution included the selection and preparation of the clinical records, manual morbidity annotation, development and refinement of the Bio_ClinicalBERT-centred pipeline, configuration and extension of the cTAKES-centred pipeline, terminology and ICD-10 mapping integration, construction and review of the evaluation data, implementation of the evaluation procedure, and analysis and interpretation of the results.

OpenAI Codex was used to assist with selected programming, debugging and technical issues explanation tasks. All AI-assisted outputs were reviewed, tested and adapted by me before inclusion. AI-assisted code and the corresponding prompts are identified and submitted in accordance with the MSc dissertation AI policy.



Signature

Date 27.08.2026

Acknowledgements

I would like to express my sincere gratitude to Professor Bouamrane for their invaluable guidance, encouragement and support throughout this project. I am particularly grateful for their suggestion to extend the work by linking extracted morbidities to standardised ICD-10 codes, which became an important and rewarding part of the dissertation. I also deeply appreciate the confidence they placed in me by entrusting me with such a meaningful and challenging project. This opportunity allowed me to gain valuable knowledge, confidence and experience, for which I am sincerely thankful.

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1 Introduction

1.1 Background and Context

Electronic health records (EHRs) contain both structured data, such as laboratory results and medication orders, and unstructured clinical narratives written by healthcare professionals. These narratives provide detailed information about patients' diagnoses, previous medical conditions, treatments and clinical progress. MIMIC-IV discharge summaries [1] are particularly valuable because they consolidate information from an entire hospital admission. However, relevant disease information may be distributed across several sections, expressed using different terminology or repeated multiple times. Consequently, reconstructing a patient's morbidity history through manual review can be time-consuming.

Natural language processing (NLP) offers transforming free-text clinical records into structured data that can be retrieved, compared and analysed more efficiently. In this project, morbidity extraction refers to the identification of diseases and pathological conditions documented as affecting the patient. This task involves more than detecting words that resemble disease names. Clinical narratives frequently contain abbreviations, spelling variations, misspellings, incomplete sentences and non-standard expressions. For example, *HTN*, *hypertension* and *high blood pressure* may all describe the same clinical concept.

Clinical context introduces an additional challenge because the occurrence of a disease term does not necessarily indicate that the patient has that disease. A mention may be negated, uncertain, hypothetical or associated with a family member. For example, *pneumonia* should be retained in "the patient was treated for pneumonia" but excluded in "*no evidence of pneumonia*". Similarly, "*the patient's mother has diabetes*" describes family history rather than a morbidity of the patient. A reliable extraction system must therefore combine morbidity detection with assertion and contextual processing.

Extracted disease descriptions also need to be standardised. SNOMED CT provides a detailed clinical terminology for representing information in electronic records [2], whereas ICD-10 is primarily a classification used to categorise diseases for reporting, statistical analysis and other secondary purposes [3]. NHS Digital distinguishes SNOMED CT as a structured clinical vocabulary from ICD-10 as a statistical classification of diagnostic information [2], [3]. Linking different clinical expressions to a common concept and subsequently to a suggested NHS ICD-10 5th Edition category can reduce ambiguity and facilitate consistent reuse of the extracted information. However, automatically generated codes should be treated as suggestions rather than replacements for formal clinical coding.

Clinical information extraction can be implemented using terminology-driven systems or contextual language models. Apache cTAKES is an open-source natural language processing (NLP) system for the clinical domain that generates linguistic and semantic annotations and supports the recognition of clinical entities and their mapping to UMLS concepts [4], [5]. Its terminology-based architecture may offer broad concept coverage and interpretable code suggestions, although its performance depends on dictionary coverage, pipeline configuration and assertion processing.

In contrast, models based on the Transformer architecture employ a different approach, learning to generate contextual language representations. The Bio_ClinicalBERT model was pre-trained on clinical texts and it can be fine-tuned for tasks such as named entity recognition in clinical notes [6]. Such models may be better able to interpret a term from the surrounding context, but they require task-specific annotated data and do not automatically provide standard medical codes. EHRKit provides a Python-based environment for applying different NLP functions and neural approaches to clinical text [7]. Terminology-driven and transformer-based

approaches have different potential strengths, and their relative suitability for document-level morbidity extraction requires direct evaluation.

This project compares a Bio_ClinicalBERT-centred pipeline implemented within EHRKit with a terminology-centred cTAKES pipeline. Both systems process the same complete MIMIC-IV discharge summaries and produce concept-level morbidity lists with suggested NHS ICD-10 5th Edition families.

The purpose of the project is to compare the extraction and coding performance of the two approaches, identify their complementary strengths and error patterns, and assess the potential value of a hybrid morbidity-extraction system.

1.2 Research questions

The project addresses the following research questions:

- 1) How accurately can the fine-tuned Bio_ClinicalBERT pipeline implemented within EHRKit and an Apache cTAKES-centred pipeline extract unique morbidity concepts from full texts of discharge summaries in the MIMIC-IV dataset?
- 2) How effectively can the morbidities extracted by each pipeline be normalised and assigned appropriate suggested NHS ICD-10 5th Edition families?
- 3) What complementary strengths, limitations and error patterns do the transformer-centred and terminology-centred approaches demonstrate, and what are the implications for the design of a hybrid morbidity-extraction system?

1.3 Scope and Objectives

The project focuses on extracting diseases and pathological conditions from complete MIMIC-IV discharge summaries. The unit of evaluation is a unique morbidity concept within an individual document. Repeated mentions, abbreviations and synonymous descriptions representing the same condition are consolidated and counted only once. Confirmed current and historical patient conditions are included, while negated, uncertain, hypothetical, conditional, family-related and non-diagnostic mentions are excluded.

The scope does not include the extraction of medications, dosages, clinical procedures or laboratory results. ICD-10 outputs are evaluated as automated suggestions and are not intended to replace coding performed by qualified clinical coders.

The study uses patient-disjoint data for three distinct purposes. A corpus of 100 manually annotated discharge summaries supports Bio_ClinicalBERT development, 60 additional records support iterative refinement of both pipelines, and 20 final held-out records are reserved for the paired comparison.

The objectives of the project were:

- To prepare complete MIMIC-IV discharge summaries and create occurrence-level development annotations and a concept-level final gold standard for morbidity extraction.
- To develop, fine-tune and refine a Bio_ClinicalBERT morbidity-extraction pipeline within the EHRKit environment.
- To configure and refine an Apache cTAKES-centred pipeline for extracting morbidities from the same type of discharge summaries.

- To apply context filtering and deduplication within a record so that both systems produce comparable lists of unique patient morbidities.
- To normalise extracted morbidities using UMLS and SNOMED CT UK resources and assign suggested NHS ICD-10 5th Edition families.
- To compare the two pipelines on the same final held-out records using precision, recall and F1 score.
- To evaluate ICD-10 coding coverage, correct-code coverage and coding accuracy, and to analyse the principal strengths and limitations of each approach.

1.4 Main contribution

The main contribution of this dissertation is a **controlled comparison of transformer-centred and terminology-centred approaches to document-level morbidity extraction**. Both complete pipelines were evaluated on the same held-out MIMIC-IV discharge summaries using a common concept-level gold standard, duplicate-exclusion policy and scoring procedure.

The project also produced a patient-disjoint annotated corpus and an auditable end-to-end workflow covering morbidity extraction, contextual filtering, concept-level deduplication, terminology normalisation and suggested NHS ICD-10 5th Edition coding. This extends the comparison beyond entity recognition by examining whether extracted morbidity information can be converted into a standardised and reusable representation.

The comparison revealed complementary and consistent system behaviour across the expanded evaluation. *The Bio_ClinicalBERT-centred pipeline achieved substantially higher precision, while the cTAKES-centred pipeline achieved higher recall and stronger ICD-10 coding performance. Although Bio_ClinicalBERT achieved a modestly higher F1 score in the expanded 200-record evaluation, neither approach was uniformly superior across the evaluated dimensions. The persistence of these contrasting error profiles at larger scale further motivates future evaluation of a hybrid pipeline combining terminology-based candidate coverage with transformer-based contextual validation.*

1.5 Overview of dissertation

Chapter 2 reviews related work on clinical concept extraction, contextual processing, transformer models, terminology normalisation and automatic medical coding, and identifies the research gap addressed by the dissertation.

Chapter 3 presents the experimental methodology, including data preparation, annotation, partitioning, refinement and evaluation design.

Chapter 4 describes the implementation of the Bio_ClinicalBERT-centred and cTAKES-centred pipelines.

Chapter 5 reports the development-stage and final held-out results.

Chapter 6 evaluates the project against its research questions and objectives, compares the findings with existing work, discusses limitations and proposes future development.

2 State-of-the-Art

Research on extracting information about morbidity from clinical texts covers several related but distinct tasks:

- Detecting disease mentions;
- Determining their clinical context;
- Normalising alternative expressions to standard concepts;
- Assigning classification codes.

Previous studies often evaluate these tasks independently. However, producing a structured morbidity summary requires them to operate as a connected pipeline. This chapter reviews the approaches most relevant to that process and identifies the gap addressed by this dissertation.

2.1 Terminology-driven clinical concept extraction

Early clinical information extraction systems relied primarily on dictionaries, medical terminologies and manually defined linguistic rules. These approaches identify text matching known clinical expressions and associate the extracted mention with a concept from a terminology such as the Unified Medical Language System (UMLS). Their main advantage is that entity recognition and concept normalisation can be performed together, producing interpretable concept identifiers without requiring a large task-specific training dataset.

Apache cTAKES is one of the most mature open-source systems in this category. It was developed as a modular natural language processing (NLP) pipeline for the clinical domain, based on the Unstructured Information Management Architecture (UIMA) and includes components for sentence segmentation, tokenisation, part-of-speech tagging, clinical entity recognition, UMLS concept mapping and assertion processing [5]. These contextual attributes allow extracted concepts to be classified as current, historical, possible, related to family history, or negated.

The terminology-driven design of cTAKES offers several advantages for morbidity extraction. It can recognise concepts from a large clinical vocabulary, provide UMLS identifiers and generate outputs that can be inspected at each processing stage. It can also be applied without first training an entity-recognition model on a large local corpus. However, its accuracy depends on the coverage of its dictionaries, the quality of lexical matching and the configuration of its processing components. Ambiguous abbreviations, misspellings and non-standard expressions may be incorrectly mapped or missed, while broad dictionary matching can generate concepts that are present in the text but do not represent confirmed patient morbidities.

Lossio-Ventura et al. compared cTAKES with CLAMP, MetaMap, NCBO Annotator, QuickUMLS and ScispaCy using i2b2 and MIMIC-III records [8]. Their results showed substantial variations across systems and evaluation conditions. For example, performance differed between exact and inexact matching and across challenges such as negation, abbreviations, ambiguity and misspelling. No system was consistently optimal for all extraction problems. This demonstrates that the suitability of a clinical NLP system depends on the specific entity definition, dataset and evaluation method.

Other systems combine terminology resources with statistical learning. MedCAT, for example, uses self-supervised learning to recognize clinical concepts and map them to standard terminologies such as UMLS or SNOMED CT. It has been evaluated on large collections of clinical records from UK hospitals [9]. This work demonstrates the potential benefit of combining contextual representations with medical terminologies. However, effective local adaptation of MedCAT may require a substantial collection of domain-specific clinical text, and *its reported*

evaluations do not directly answer how cTAKES compares with a task-specific Bio_ClinicalBERT pipeline for unique morbidity extraction from complete MIMIC-IV discharge summaries.

2.2 Assertion and contextual processing

Clinical entity recognition alone is insufficient for constructing a morbidity history. A document may mention a condition that is denied, is hypothetical or uncertain in nature, relates to the past, or concerns another person rather than the patient. Systems that do not process these properties may achieve high mention coverage while generating clinically misleading false positives.

NegEx introduces a rule-based method in which lexical triggers such as ‘no’, ‘denies’ and ‘ruled out’ determine whether a nearby clinical concept is negated [10]. ConText extends this approach to additional contextual properties, including temporality, hypothetical status and experiencer [16]. Harkema et al. found that surface-level contextual rules could achieve useful performance across several report types. However, they also concluded that some distinctions, particularly between recent and historical conditions, require information beyond local lexical triggers.

Previous evaluation studies have generally measured assertion processing at the individual mention level. **The present project instead requires contextual decisions to contribute to a document-level list of unique patient morbidities.** This introduces an additional aggregation problem because different mentions of the same disease may have different contextual properties across a document. For example, a condition appearing in a Family History or Discharge Instructions section does not necessarily represent a confirmed morbidity of the patient.

2.3 Transformer-based clinical concept extraction

The introduction of transformer models changed clinical concept extraction by enabling systems to learn context-dependent representations instead of relying exclusively on dictionary matching. **BioBERT** demonstrated that continuing the pre-training of the BERT model on biomedical publications improves performance on named entity recognition, relation extraction, and question-answering tasks in the biomedical domain. [11]. However, biomedical publications differ from clinical records in their vocabulary, structure and writing style.

Bio_ClinicalBERT addressed this domain difference by adapting a biomedical BERT model to clinical notes. Alsentzer et al. released models trained on general clinical text and discharge summaries and found that clinical-domain pretraining improved performance on several, although not all, evaluated clinical NLP tasks [6]. These findings indicate that domain-specific pretraining is beneficial but does not remove the need for task-specific fine-tuning and validation.

Nevertheless, benchmark performance does not transfer automatically to document-level morbidity extraction. Public clinical NLP datasets use different entity definitions, document types and annotation policies. Many evaluate individual spans rather than unique concepts aggregated across a complete record. Transformer models also produce entity spans rather than standard terminology identifiers, meaning that concept normalisation must be performed separately.

Complete discharge summaries present an additional technical challenge because their length commonly exceeds the input limit of BERT-based models. The text must therefore be divided into sections or overlapping chunks. Pascual et al. similarly identified long-document processing as a principal limitation of BERT-based automatic ICD coding [12]. Chunking can separate a disease mention from relevant contextual information and can create duplicate predictions in overlapping text segments.

EHRKit provides a Python environment that integrates multiple NLP libraries and supports a range of EHR-processing tasks [7]. It should be distinguished from an extraction model: EHRKit provides the environment in which a model can be implemented, whereas extraction performance is determined by the selected model, its training data and the surrounding processing rules. Consequently, **the comparison in this dissertation is specifically between a fine-tuned Bio_ClinicalBERT pipeline implemented within EHRKit and a configured cTAKES pipeline**, rather than between EHRKit and cTAKES as equivalent algorithms.

2.4 Concept normalisation and medical coding

Clinical concept normalisation maps different textual expressions to a common medical concept. *Terminology-driven systems such as cTAKES perform this operation by associating matched expressions with UMLS Concept Unique Identifiers. Transformer-based entity-recognition models do not normally provide this link directly; thus, a separate normalisation stage is required.*

Concept normalisation should be distinguished from automatic ICD coding. In most studies on automated coding, the assignment of ICD codes is treated as a multi-label document classification task. For example, Mullenbach et al. developed the CAML architecture, which uses label-specific attention to predict ICD codes from complete clinical notes and highlight relevant text for each prediction [13]. This approach predicts the collection of codes associated with the document but does not necessarily extract and validate each individual morbidity mention before coding it.

Transformer-based coding methods face difficulties associated with long documents, large label spaces and rare codes, meaning that they must select from a large number of possible ICD codes, while many uncommon codes are represented by only a small number of examples in the training data. Pascual et al. found that pre-trained transformers could achieve competitive results using selected portions of clinical documents but that aggregating information across long records remained a major limitation [12]. *This highlights the continuing need for controlled evaluation using manually reviewed clinical data.*

Mapping from SNOMED CT to ICD-10 is also not a simple one-to-one lookup. The two systems serve different purposes and represent information at different levels of detail. NHS guidance states that national cross-maps are only semi-automated because additional information from the EHR may be required before a final classification code can be assigned [2]. SNOMED International similarly uses complex and extended map reference sets when contextual rules are required to determine an ICD-10 target [15].

Therefore, coding evaluation should assess:

- **code-assignment coverage** — the proportion of correctly extracted morbidities that receive an ICD-10 suggestion;
- **coding accuracy** — the proportion of suggestions assigned to correctly extracted morbidities that are correct; and
- **correct-code coverage** — the proportion of correctly extracted morbidities that receive the correct ICD-10 suggestion.

These distinctions motivate the use of coding coverage, correct-code coverage and coding accuracy in addition to precision, recall and F1.

2.5 Comparison of related work

Table 2.1 Comparison of clinical NLP approaches

Study or system	Approach	Main evaluation focus	Limitation relative to this project
Savova et al. [5]	Modular terminology-driven cTAKES pipeline	Clinical entity recognition and linguistic annotation	Does not provide a direct comparison with a fine-tuned clinical transformer on MIMIC-IV
Harkema et al. [16]	Rule-based contextual processing	Negation, temporality, hypothetical status and experienter	Evaluates contextual properties separately from end-to-end morbidity normalisation and coding
Lossio-Ventura et al. [8]	External comparison of six concept-recognition systems	Multiple entity types and extraction challenges on i2b2 and MIMIC-III	Uses mention-level evaluation and does not compare cTAKES with a task-specific Bio_ClinicalBERT model
Kraljevic et al. [9]	Self-supervised concept recognition with UMLS/SNOMED CT	Clinical concept extraction across large hospital datasets	Requires substantial local training data and does not evaluate the specific cTAKES - ClinicalBERT comparison
Alsentzer et al. [6]	Clinical-domain BERT pretraining	General clinical NLP benchmark tasks	Does not evaluate unique morbidity aggregation or NHS ICD-10 mapping
Mullenbach et al. [13] and Pascual et al. [12]	Document-level automatic ICD coding	Multi-label prediction of codes from clinical notes	Predict codes directly rather than extracting, validating and deduplicating individual morbidities
Falis et al. [14]	Generative-model coding and data augmentation	ICD-10 coding of synthetic and real MIMIC-IV summaries	Limited reliability on real records and no task-specific morbidity gold standard

These studies are not directly comparable through their reported F1 scores because they use different corpora, annotation schemes, entity categories and matching criteria. Table 2.1 should therefore be used to compare research design and task coverage rather than to rank the systems quantitatively.

2.6 Identified Research Gap

Existing research demonstrates that terminology-driven systems can provide broad concept coverage and direct links to standard vocabularies, while transformer models can learn contextual patterns that are difficult to express through dictionaries and fixed rules. However, previous evaluations generally focus on one component of the problem: mention-level entity recognition, assertion classification, concept normalisation or document-level ICD prediction.

The reviewed literature does not establish how a fine-tuned Bio_ClinicalBERT morbidity-extraction pipeline compares with cTAKES when both systems are:

- applied to the same complete MIMIC-IV discharge summaries;
- evaluated using the same manually created gold standard;
- assessed at the level of unique morbidity concepts within each document;
- required to exclude mentions according to clinical context;
- evaluated for both morbidity extraction and suggested NHS ICD-10 coding.

This dissertation addresses that gap through a controlled comparison of the two approaches on the same previously unseen records. The evaluation separates extraction performance from coding performance and measures precision, recall, F1 score, ICD-10 coding coverage, correct-code coverage and coding accuracy. It therefore examines not only whether the systems can detect disease expressions, but whether they can produce a reliable and standardised summary of a patient's morbidities.

The comparison also examines whether the two approaches produce complementary errors. This is important because similar aggregate performance can conceal different operating behaviour: one system may prioritise precision, while another may provide greater recall or terminology coverage. Identifying such trade-offs provides a basis for assessing whether a hybrid approach could combine the strengths of both systems.

3 Experimental Methodology

This chapter presents the methodology and experimental controls used to support a reproducible and fair comparison of the two morbidity-extraction pipelines. Their technical implementation and performance results are presented in Chapters 4 and 5, respectively.

3.1 Research design

This study uses a **controlled, paired comparison of two complete NLP pipelines for converting free-text discharge summaries into structured lists of patient morbidities**.

The first was a transformer-centred pipeline based on a fine-tuned **Bio_ClinicalBERT** sequence tagger [6] implemented within the EHRKit environment [7].

The second was a terminology-centred pipeline based on **Apache cTAKES** [5] and its UMLS-linked DiseaseDisorderMention annotations [4], [5]. The study evaluates the complete implemented workflows rather than the unmodified Bio_ClinicalBERT and cTAKES systems in isolation.

Development was iterative: supervised Bio_ClinicalBERT training and model selection were followed by error-driven refinement of the contextual, terminology, deduplication and mapping components of both workflows. *The refinement procedure is described in Section 3.5.*

The experiment consisted of three sequential phases. In **the first phase**, a patient-disjoint corpus of 100 manually annotated MIMIC-IV discharge summaries was used for Bio_ClinicalBERT development [1]. It was divided into 70 training notes, 10 calibration notes and 20 initial internal-validation notes. The training partition was used to optimise the model weights, while the calibration partition supported checkpoint and confidence-threshold selection. The initial internal-validation partition was used to assess the first complete version of the transformer-centred pipeline. cTAKES did not require supervised training on these annotations because its primary extraction process was based on terminology resources and assertion-processing components.

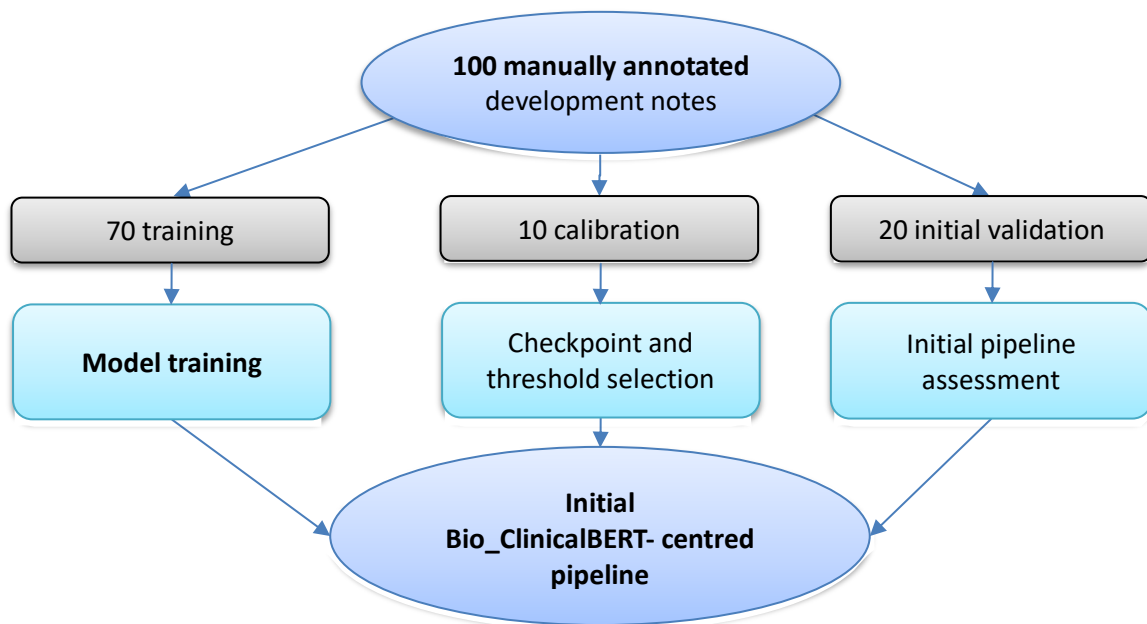


Figure 3.1 **PHASE 1: BIO_CLINICALBERT development**

The second phase used 60 additional discharge summaries from patients not represented in the first-phase partitions. The records supported manual error analysis and iterative refinement of both complete pipelines. Because their outputs informed subsequent development decisions, these records were treated as development data and were excluded from the final evaluation.

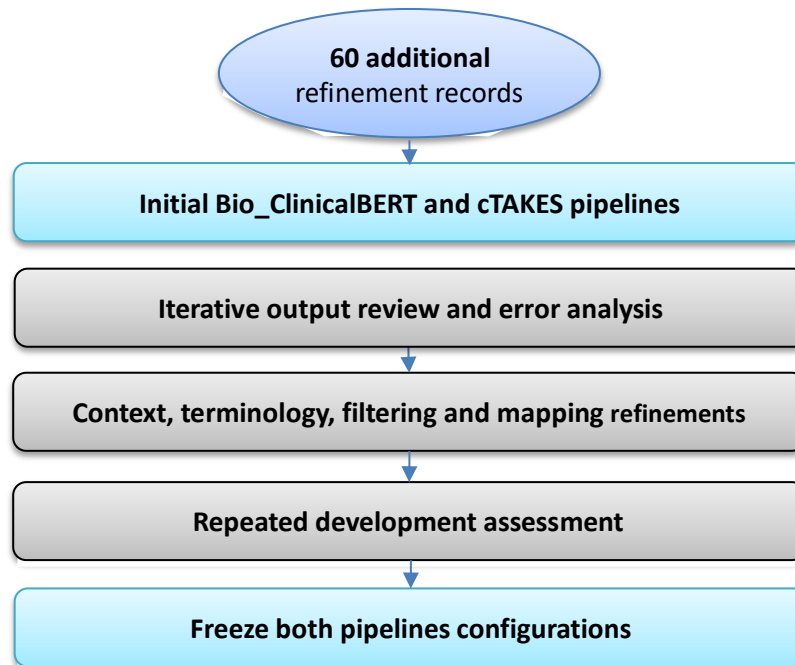


Figure 3.2 **PHASE 2: Pipeline refinement**

In **the third phase**, both pipeline configurations were frozen and evaluated on a final set of 20 discharge summaries from patients not represented in any development partition.

These records were not used for model training, checkpoint selection, threshold calibration, error analysis or rule development. Both pipelines processed the same clinical texts, and their outputs were compared with the same manually reviewed gold standard using identical concept-matching, duplicate-exclusion and scoring rules.

The final unit of evaluation was a unique morbidity concept within a note rather than each individual textual occurrence of a disease. Repeated mentions and synonymous expressions representing the same condition were consolidated before scoring. This prevented frequently repeated conditions from having a disproportionate effect on the results.

Figure 3.1, Figure 3.2, Figure 3.3 illustrate the separation between supervised Bio_ClinicalBERT development, iterative refinement of both end-to-end pipelines and final held-out evaluation. Patient and record separation across the three phases reduced the risk of information leakage and ensured that the final results represented the performance of frozen pipeline configurations on previously unused data.

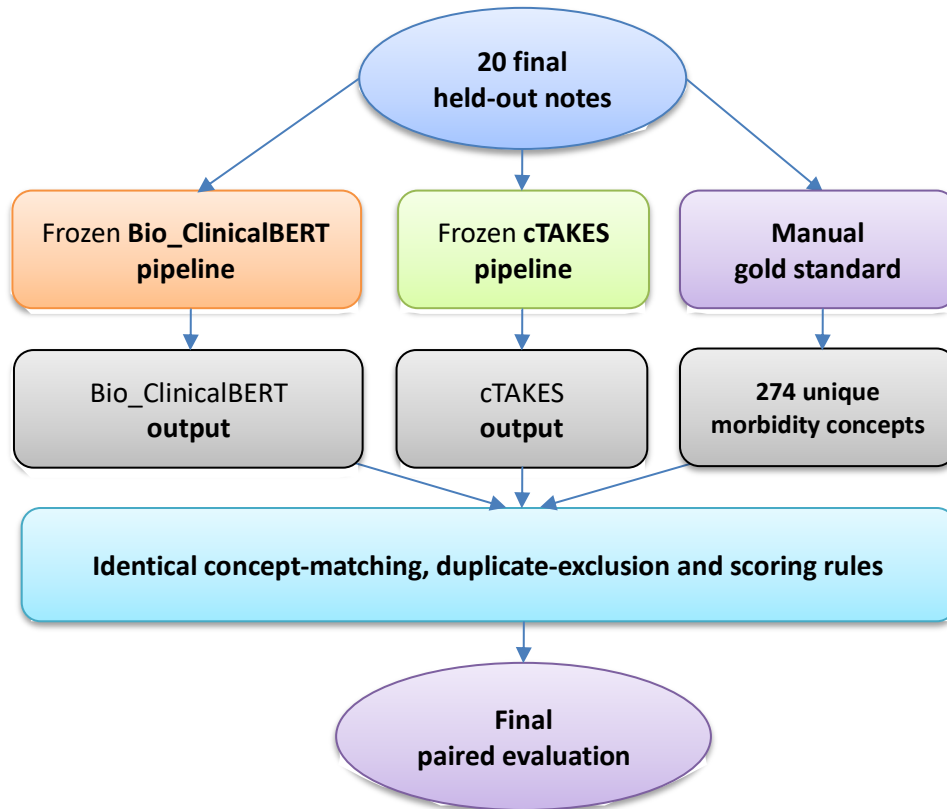


Figure 3.3 PHASE 3: Final paired evaluation

3.2 MIMIC-IV Dataset and records selection

The clinical text used in this project was obtained from MIMIC-IV-Note, a de-identified collection of free-text clinical documentation linked to the MIMIC-IV database [1]. Complete *discharge summaries* were selected because they provide a consolidated account of a hospital admission, including diagnoses, previous medical history, investigations, treatment and discharge planning. However, their length, repeated information, abbreviations and variation in section structure make them a challenging source for automated morbidity extraction. MIMIC-IV-Note contains de-identified clinical data. Access was obtained through PhysioNet under its credentialing and data-use requirements, and only the de-identified database identifiers were used during processing [1].

The study used 180 discharge summaries across the three experimental phases described in [Section 3.1](#). The first **100 notes formed the Bio_ClinicalBERT development corpus**, **60 additional notes were used for iterative refinement of both pipelines**, and **20 further notes formed the final held-out evaluation set**. Each selected note was associated with a note identifier, subject identifier and hospital-admission identifier.

Patient-level separation was applied to prevent notes belonging to the same patient from appearing in different experimental partitions. Where multiple eligible records were available for one patient, only one discharge summary was retained. The final evaluation records were selected only after excluding note, patient and admission identifiers represented in the development and refinement data.

The complete text of each selected discharge summary was retained, including its clinical sections and line structure. During the final evaluation, both pipelines received the same discharge-summary text and record identifiers. CSV files were used as the final cTAKES input,

while Excel workbooks were used for structured review and evaluation. These formatting differences did not alter the clinical information available to either pipeline.

3.3 Manual annotation and gold-standard construction

3.3.1 Bio_ClinicalBERT development annotations

The Bio_ClinicalBERT development corpus contained 100 manually annotated discharge summaries. The notes were uploaded to Label Studio, where words and phrases referring to diseases or pathological conditions were highlighted and assigned the MORBIDITY label. After annotation, the labelled text spans, character positions and record identifiers were exported from Label Studio for data preparation and supervised model development.

The occurrence-level annotations were subsequently converted into BIO sequence labels. The beginning of a morbidity expression was represented as *B-MORBIDITY*, subsequent tokens within the same expression as *I-MORBIDITY*, and all other tokens as *O*. This representation allowed multi-word expressions such as *chronic kidney disease* to be learned as a single morbidity span.

The occurrence-level Label Studio annotations were used for model development and should be distinguished from the concept-level gold standard used for the final evaluation. The training annotations identified textual spans, whereas the final evaluation assessed one unique morbidity concept per note.

3.3.2 Annotation quality control

The initial Label Studio export contained **1,927 manually annotated morbidity spans**. A quality-control procedure was applied before the spans were converted into BIO labels. For **47 annotations**, the text value stored by Label Studio did not exactly match the corresponding substring at the recorded character positions. In these cases, the source discharge-summary text at the recorded positions was treated as authoritative.

Five exact duplicate spans were removed. A further **11 annotations** belonged to overlapping or nested groups that could not be represented unambiguously using a single BIO label sequence. In these cases, the longest clinically meaningful span was retained. Following these checks, 1,911 BIO-compatible morbidity spans remained for Bio_ClinicalBERT development.

These checks reduced avoidable label noise. Duplicate spans could cause the same phrase to be represented more than once, while incompatible overlapping spans could assign conflicting labels to the same tokens.

3.3.3 Operational definition of morbidity

For the final structured output, morbidity was defined as a disease or pathological condition affirmed for the patient. Both current and confirmed historical conditions were eligible for inclusion.

A disease mention was excluded from the primary morbidity output when it was:

- negated;
- uncertain, suspected or presented as a differential diagnosis;
- hypothetical or conditional;
- attributed to a family member or another person;
- included only in patient instructions or educational material; or
- present only in a non-diagnostic context, such as an allergy or medication list.

Medications, procedures, isolated anatomical structures and laboratory measurements were outside the scope of morbidity extraction. Symptoms and clinical findings were excluded unless they constituted a diagnosed pathological condition. Confirmed injuries, including fractures and haemorrhages, were retained when they satisfied the project’s morbidity definition. Other clinical events could be retained in audit outputs but were not included in the primary disease-oriented evaluation.

This distinction was important because **detecting a disease term did not automatically establish that the patient had that disease**. For example, *pneumonia* in *no evidence of pneumonia* was a valid candidate morbidity mention, but it was not an eligible patient morbidity after contextual classification.

3.3.4 Refinement records and final gold standard

The **60 records** used during the second experimental phase supported manual output review and error analysis. Predictions from both pipelines were inspected to identify recurrent extraction, context, terminology and coding errors. Because these records influenced pipeline-development decisions, they were treated as development data and were not included in the final performance evaluation.

The gold standard for the **final 20 held-out records contained 274 unique morbidity concepts**. The final reference standard was manually constructed through direct review of the 20 discharge summaries before the system outputs were scored. Eligible patient morbidities were recorded according to the definition in Section 3.3.3 and consolidated at concept level within each note. The fixed reference list was subsequently used to review the predictions from both pipelines. Repeated textual occurrences of the same disease within a note were counted only once. Synonymous or abbreviated expressions were also consolidated when they represented the same clinical condition. For example, *DM2* and *type 2 diabetes mellitus* could be treated as one morbidity concept when the surrounding clinical context supported the same diagnosis.

Concept-level consolidation **prevented frequently repeated conditions** from having a disproportionate effect on the evaluation. It also aligned the gold standard with the intended system output: a concise list of distinct patient morbidities rather than a list of every textual mention.

3.4 Data partitioning and leakage control

Table 3.1 summarises the experimental partitions and their roles.

Table 3.1 Experimental data partitions and their roles

Partition	Records	Purpose	Included in final evaluation
Bio_ClinicalBERT training	70	Optimisation of model weights	No
Calibration	10	Checkpoint and confidence-threshold selection	No
Initial internal validation	20	Assessment of the initial transformer-centred pipeline	No
Pipeline refinement	60	Error analysis and iterative refinement of both pipelines	No
Final held-out evaluation	20	Paired evaluation of the frozen pipelines	Yes

The **100-note** development corpus was divided into **70 training** notes, **10 calibration** notes and **20 initial internal-validation** notes. Before partitioning, the data-loading procedure verified that the note, subject and admission identifiers were unique and rejected duplicated identifiers. Each discharge summary in the development corpus therefore represented a different patient and hospital admission.

Partition assignment was based on *SHA-256 values* calculated from a fixed split seed together with each record's subject and note identifiers. The records were ranked by these values, making the split reproducible and independent of their original order in the source file. *Twenty records were first assigned to internal validation. The remaining records were ranked again using a calibration-specific version of the split key; 10 were assigned to calibration and the remaining 70 to training.* Partition sizes, calibration and validation identifiers, and source-data information were recorded in the experimental manifest. The training partition was reproducibly defined by the remaining records.

The training partition was used to optimise the Bio_ClinicalBERT model weights. A random seed of 42 was fixed for Python, NumPy and PyTorch operations to improve the reproducibility of stochastic model training. The calibration partition was used to select the model checkpoint and minimum entity-confidence threshold. The 20-note internal-validation partition was then used to assess the first complete version of the transformer-centred pipeline.

Because **the internal-validation results subsequently informed further development**, this partition was not treated as the final independent test set. The **60 additional refinement records** were also classified as development data because their outputs were reviewed and used to modify the two pipelines. The refinement procedure is described in [Section 3.5](#).

Only the **final 20 records were reserved for the paired evaluation** reported as the principal result of the dissertation. After all model-selection and pipeline-refinement decisions had been completed, both configurations were frozen. The final records were not used to retrain the model, select a checkpoint, adjust the confidence threshold, modify contextual or terminology rules, or select ICD-10 mappings.

Patient-level separation was enforced across the experimental partitions using subject identifiers. In this study, **patient-disjoint** means that a patient represented in one partition was not represented in another. This separation reduced the risk of data leakage caused by the same patient's clinical history or recurring documentation patterns appearing during both pipeline development and final evaluation.

3.5 Iterative pipeline-refinement procedure

Following the initial internal assessment, pipeline refinement was conducted as an error-driven and iterative development process. **Both systems processed the 60 refinement records**, after which their structured outputs were manually reviewed. *Recurrent false-positive, false-negative, normalisation and coding errors were identified and grouped according to their likely causes.* The relevant processing components were then modified, and the pipelines were reassessed on development data. This **review–refinement–reassessment cycle was repeated** during the development phase.

For the Bio_ClinicalBERT-centred pipeline, refinement focused on:

- contextual filtering of negated, uncertain and non-patient mentions;
- confidence-based selection of predicted morbidity spans;
- section-aware interpretation of clinical context;

- linking surface expressions to standard terminology concepts;
- consolidation of synonymous and repeated disease mentions;
- removal of non-disease concepts from the primary morbidity output.

These changes affected the complete transformer-centred pipeline rather than the Bio_ClinicalBERT model alone. Some improvements resulted from post-processing, contextual analysis, terminology linking and deduplication applied after neural span prediction.

*For the **ctAKES-centred pipeline**, refinement focused on:*

- additional assertion-processing safeguards;
- recognition and contextual disambiguation of clinical abbreviations;
- restriction to appropriate UMLS semantic types;
- removal of overly broad terminology matches;
- terminology supplementation for disease expressions missed by the initial configuration;
- selection between multiple UMLS concepts associated with the same mention;
- concept-level deduplication.

The shared **normalisation and coding stages were also reviewed**. Refinements addressed unresolved disease expressions, ambiguous SNOMED CT candidates, mappings requiring additional clinical detail and cases in which several potential ICD-10 families were available. When the terminology evidence supported a single reliable NHS ICD-10 family, it was recorded as the primary suggestion. When a unique family could not be determined safely, the pipeline retained a ranked list of up to five candidate three-character ICD-10 families for manual review rather than presenting any candidate as a definitive code. Candidate families were ranked using the level of support across the linked SNOMED CT concepts, lexical similarity to the official ICD-10 family title and the ordering information contained in the official mapping resource.

The refinements addressed recurring categories of error observed in development data rather than individual examples from the final evaluation records. No final held-out record was used to introduce a new abbreviation, terminology rule, context pattern or mapping decision. Once the planned refinement experiments had been completed, **both pipeline configurations were frozen before the final 20 records were processed**.

Development-stage performance is reported separately in Chapter 5 to document the motivation and effect of the refinement process. These development results are not interpreted as independent test performance because the records influenced subsequent pipeline decisions. The principal comparison is therefore based only on the final held-out evaluation set.

3.6 Evaluation design

3.6.1 Evaluation and matching rules

The final evaluation was conducted at the level of a unique morbidity concept within each discharge summary. A system prediction **was compared with the gold standard** for the same note. Matching was concept-based rather than being restricted to identical character strings.

A prediction was accepted as a match when it represented the same clinical condition as a gold-standard morbidity, including an accepted synonym or abbreviation. Clinically distinct subtypes were not merged. For example, *type 1 and type 2 diabetes*, acute and chronic conditions,

or *left-* and *right-sided disorders* were treated as separate concepts where the distinction was clinically relevant.

Repeated predictions representing the same morbidity within one note **were excluded as duplicates**. Consequently, each system could receive a maximum of one true positive for a particular morbidity concept in a particular note.

A prediction was classified as:

- **true positive (TP)** when it matched an eligible gold-standard morbidity;
- **false positive (FP)** when it did not correspond to an eligible gold-standard morbidity;
- **false negative (FN)** when a gold-standard morbidity was not identified by the system.

True negatives (TN) were not calculated because the number of all possible diseases not mentioned in a clinical record cannot be meaningfully enumerated.

3.6.2 Morbidity-extraction metrics

Precision measures the proportion of correctly extracted morbidity concepts:

$$Precision = \frac{TP}{TP + FP}.$$

Recall measures the proportion of identified gold-standard morbidity concepts:

$$Recall = \frac{TP}{TP + FN}.$$

The **F1** score measures the balance between precision and recall:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}.$$

Precision, recall and F1 metrics were calculated by summing TP, FP and FN count across all 20 final records.

Precision is particularly relevant when false diagnoses in a structured patient summary must be minimised.

Recall represents the completeness of the resulting morbidity list.

F1 provides a single summary measure that balances these two properties, although it does not indicate whether errors are primarily false positives or false negatives.

3.6.3 ICD-10 coding metrics

Coding performance was evaluated only for correctly extracted morbidity concepts. Three related measures were calculated.

ICD-10 coding coverage measured the proportion of true-positive morbidity concepts that received an ICD-10 suggestion:

$$Coding\ coverage = \frac{TP\ morbidities\ assigned\ a\ code}{all\ TP\ morbidities}.$$

ICD-10 coding accuracy measured the proportion of assigned code suggestions that were correct:

$$\text{Coding accuracy} = \frac{\text{TP morbidities assigned the correct code}}{\text{TP morbidities assigned any code}}.$$

Correct-code coverage measured the proportion of all correctly extracted morbidity concepts that also received the correct code:

$$\text{Correct-code coverage} = \frac{\text{TP morbidities assigned the correct code}}{\text{all TP morbidities}}.$$

These measures answer different questions. *Coding coverage* indicates how frequently the pipeline provides a suggestion. *Coding accuracy* indicates how reliable the suggestions are when a code is provided. *Correct-code coverage* captures the end-to-end proportion of correctly extracted morbidities that are also correctly coded.

The evaluation was performed at the level of suggested three-character NHS ICD-10 5th Edition families. Where a pipeline returned more than one candidate family, the candidate set was marked as correct if it contained an acceptable three-character ICD-10 family.

Table 3.2 Interpretation of the evaluation metrics

Metric	Main question
Precision	How many extracted morbidities were correct?
Recall	How many gold-standard morbidities were found?
F1	What was the balance between precision and recall?
ICD-10 coding coverage	How many correctly extracted morbidities received a code?
ICD-10 coding accuracy	How many assigned codes were correct?
ICD-10 correct-code coverage	How many correctly extracted morbidities received the correct code?

3.6.4 Evaluation implementation

The predictions from both pipelines were converted into a *common review format containing the note identifier, extracted morbidity, normalised concept, terminology identifiers where available and suggested ICD-10 family*. Manual review decisions identified accepted concept matches, false-positive predictions and duplicate rows.

In the evaluation workbook, a yellow morbidity cell represented an accepted true-positive concept, while a grey morbidity cell represented a duplicate excluded from scoring. For true-positive rows, a non-empty ICD-10 cell indicated that at least one suggestion had been provided, and a yellow ICD-10 cell indicated that the suggestion had been judged correct. The evaluation program aggregated these recorded decisions but did not independently determine clinical equivalence or coding correctness.

The evaluation program then aggregated the review decisions across the 20 final records. Gold-standard concepts not represented among the accepted predictions were counted as false negatives. The same aggregation and scoring rules were applied to both systems.

The **comparison was descriptive and paired** because both pipelines processed the same records and were assessed against the same gold standard. Consequently, small numerical differences—particularly the 0.33-percentage-point difference between the final F1 scores—should not be interpreted as evidence of statistically significant superiority.

4 System design and implementation

This chapter describes the technical implementation of two end-to-end pipelines for extracting morbidity entities. The first pipeline combined Bio_ClinicalBERT with contextual and terminology-based post-processing within EHRKit environment, while the second combined Apache cTAKES with additional assertion, terminology and mapping components.

4.1 Overall system architecture

Both pipelines received the same discharge-summary text and produced a comparable structured output. However, their internal extraction mechanisms differed. The transformer-centred pipeline used a fine-tuned Bio_ClinicalBERT model to identify morbidity spans from their linguistic context. The terminology-centred pipeline used cTAKES dictionary lookup, UMLS concepts and assertion attributes to identify candidate diseases.

After candidate extraction, both pipelines applied context processing, semantic filtering, concept normalisation, within-note deduplication and suggested NHS ICD-10 coding. The final outputs were converted into a common evaluation format containing one row for each unique morbidity concept identified within a note.

The **two implementations used different programming environments**. The Bio_ClinicalBERT-centred pipeline was implemented in Python within the EHRKit project environment, while the cTAKES-centred pipeline used Java and the Apache UIMA architecture. During final cTAKES execution, discharge-summary text and identifiers were supplied through CSV files, each note was processed as a temporary plain-text document, and the resulting annotations were exported as XML for Java post-processing. Excel workbooks were used for structured result review and evaluation. These implementation differences did not affect the comparison because both pipelines processed the same clinical content and were assessed using identical rules.

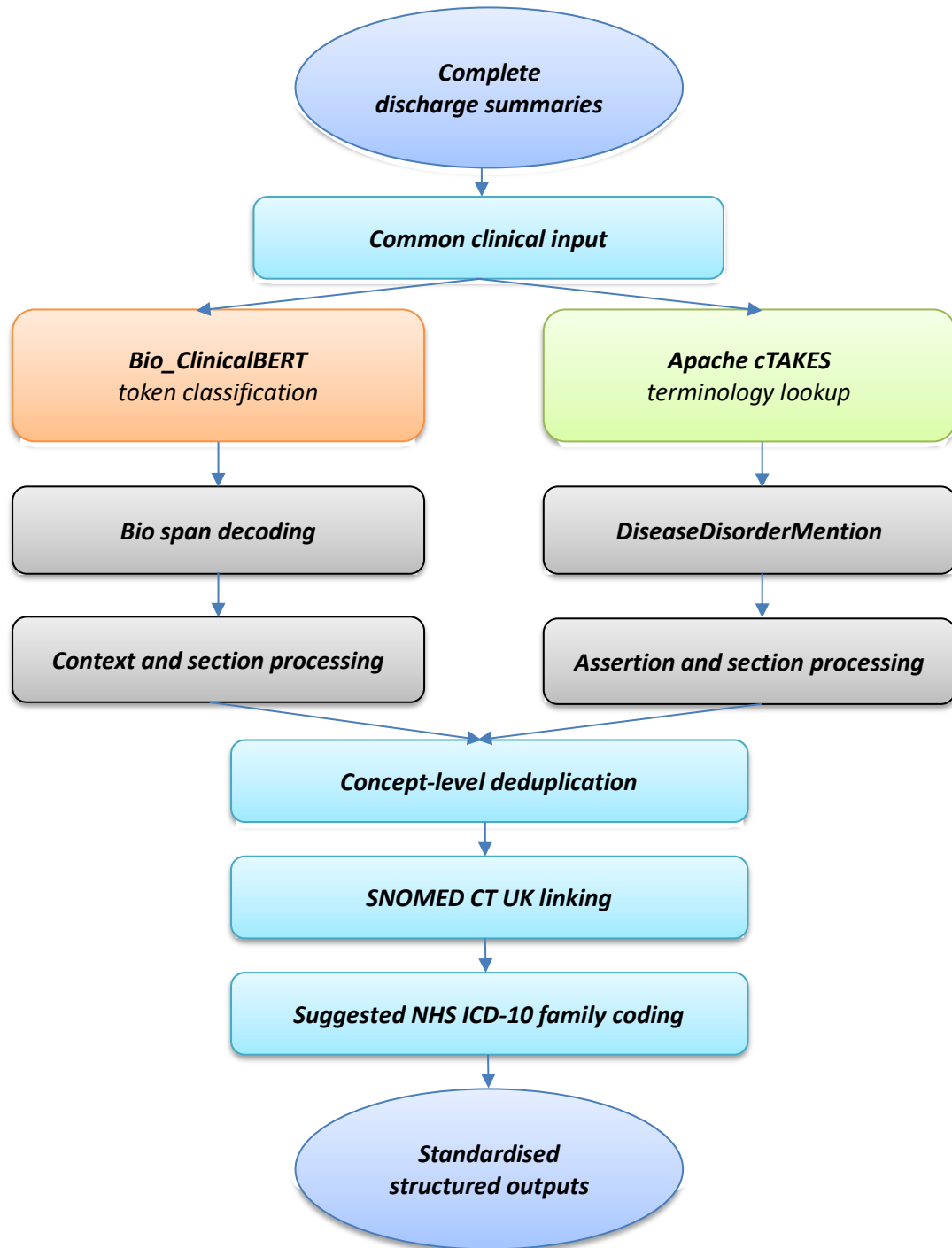


Figure 4.1 Architecture of the implemented morbidity-extraction and coding pipelines

4.2 Bio_ClinicalBERT-centred pipeline

4.2.1 Model architecture

Bio_ClinicalBERT was selected because its encoder had been pre-trained on clinical text, including MIMIC clinical notes [6]. The model was used within the EHRKit development environment [7] and adapted for token classification.

The implementation consisted of the pre-trained Bio_ClinicalBERT encoder, a dropout layer and a linear classification layer. For each input token, the classifier produced probabilities for three labels:

- O — outside a morbidity expression;
- B-MORBIDITY — the beginning of a morbidity expression;
- I-MORBIDITY — the continuation of a morbidity expression.

This design allowed the system to identify both single-word diseases and multi-word expressions such as chronic kidney disease or congestive heart failure.

Bio_ClinicalBERT was responsible primarily for predicting textual morbidity spans. Decisions concerning negation, uncertainty, patient experienter, semantic type and coding were performed by subsequent components. The final Bio_ClinicalBERT result therefore represents the complete transformer-centred pipeline rather than the neural model in isolation.

4.2.2 Tokenisation and long-document processing

The manually annotated character spans were aligned with tokens produced by the Bio_ClinicalBERT WordPiece tokenizer. Only the first WordPiece of each original word received a supervised BIO label. Additional subword pieces were assigned an ignored value and did not contribute to the training loss. This prevented long medical terms that were divided into several WordPieces from receiving disproportionate weight during training.

Complete discharge summaries frequently exceeded the transformer input limit. Each note was therefore divided into overlapping windows of 256 tokens with a stride of 64 tokens. The overlap allowed a morbidity expression located near the end of one window to be represented again in the following window.

At inference, probability vectors produced for the same word in overlapping windows were averaged. The resulting word-level probabilities were decoded using a constrained Viterbi procedure. This permitted only valid BIO transitions: an I-MORBIDITY label could follow B-MORBIDITY or another I-MORBIDITY, but could not begin a new entity or directly follow O.

Consecutive B-MORBIDITY and I-MORBIDITY predictions were combined into complete character-level spans. Span boundaries were then adjusted to full-word boundaries, and predictions with implausibly short, excessively long or otherwise invalid boundaries were rejected.

4.2.3 Fine-tuning and model selection

The Bio_ClinicalBERT model was fine-tuned on the 70-note training partition described in [Chapter 3](#). The training partition initially contained 1,173 manually annotated spans. Exact case-insensitive repetitions of an annotated phrase within the same training note were propagated as training-only weak supervision, increasing the number of training spans to 1,441. This operation was not applied to calibration, internal-validation or final evaluation records.

Table 4.1 presents the principal training parameters.

Table 4.1 Bio_ClinicalBERT training configuration

Parameter	Value
Base encoder	Bio_ClinicalBERT
Output labels	O, B-MORBIDITY, I-MORBIDITY
Maximum sequence length	256 tokens

Window stride	64 tokens
Batch size	2
Maximum epochs	8
Classifier learning rate	5×10^{-5}
Encoder learning rate	1×10^{-5}
Optimiser	AdamW
Weight decay	0.01
Linear warm-up	10% of training steps
Gradient clipping	1.0
Random seed	42
Selected epoch	7
Selected entity threshold	0.89

The model was trained for eight epochs and AdamW optimisation was used. The classifier used a learning rate five times larger than that of the encoder because its parameters were newly initialised for the morbidity task.

The checkpoint was selected using the 10-note calibration partition. Epoch 7 achieved the highest calibration selection score and was retained. Entity-confidence thresholds from 0.30 to 0.95 were then examined in increments of 0.01. The primary selection criterion was concept-level F1; ties were resolved by preferring higher recall and then the lower threshold. The selected minimum entity-confidence threshold was 0.89. *The 60 pipeline-refinement records were not used to update the neural model weights.* The **final held-out evaluation used the epoch-7 checkpoint trained on the 70-note training partition and selected using the 10-note calibration partition.** The additional records informed only the subsequent refinement of the complete processing workflow

An entity score was calculated from the token probabilities within its decoded span. Predictions below the selected threshold were excluded. The relatively high threshold made the model more conservative and contributed to the high precision observed in the final evaluation.

4.2.4 Contextual classification

Predicted morbidity spans were passed to medSpaCy ConText [17], which implements contextual processing derived from the ConText approach [16]. The component supplied indicators for properties such as negation, uncertainty, historical status and experienter.

Additional section-aware rules were applied because complete discharge summaries contain headings, lists and hard-line breaks that can cause contextual triggers to extend beyond their intended scope. Modifier cues were therefore constrained to the current clinical section and local clause. For example, denies in a Review of Systems section was not allowed to negate a diagnosis appearing later in the discharge summary.

Each candidate mention was assigned one of the following contextual classes:

- confirmed current;
- confirmed historical;

- suspected or possible;
- negated;
- family history;
- instruction or education;
- non-diagnostic context.

Only confirmed current and confirmed historical conditions attributed to the patient were eligible for the primary morbidity output. Negated, uncertain, hypothetical, family-related and instruction-only mentions remained available in the audit output but were excluded from the evaluated disease list.

This stage is important when interpreting the results. Bio_ClinicalBERT proposed the textual spans, but the complete pipeline—including medSpaCy ConText, section rules and semantic filters—determined whether a predicted span represented an eligible patient morbidity.

4.2.5 Semantic filtering and post-processing

A terminology-supported semantic profile was constructed for each contextually eligible mention. UMLS semantic types and SNOMED CT semantic tags were used to distinguish diseases from symptoms, findings, injuries, procedures, medications and anatomical structures [4].

Only disease-oriented concepts were included in the primary patient morbidities output. Confirmed symptoms, findings, injuries and other events could be retained in separate audit sheets but were not counted in the primary disease-only evaluation.

Repeated predictions were deduplicated within each note. Where a SNOMED CT identifier was available, it formed the principal deduplication key. Otherwise, the normalised morbidity expression was used. When several mentions represented the same concept, the pipeline retained a representative row using contextual status, clinical-section priority, confidence and mention length while preserving the number of merged occurrences for audit purposes.

4.3 cTAKES-centred pipeline

4.3.1 Pipeline execution and XMI processing

The terminology-centred implementation used Apache cTAKES 6.0.0 [5]. During the final evaluation, the Java orchestration program used a CSV file containing the note, subject and admission identifiers together with the complete discharge-summary text. Each note was converted into a temporary plain-text file and processed using the local cTAKES clinical pipeline and UMLS terminology configuration. The resulting cTAKES annotations were exported as XMI files for subsequent parsing and post-processing by the Java pipeline.

cTAKES produced an XMI representation of each processed note. The Java implementation parsed these XMI files and extracted DiseaseDisorderMention annotations together with their character positions, preferred terms, UMLS Concept Unique Identifiers and semantic types [4].

The local pipeline also provided assertion fields including:

- polarity;
- uncertainty;
- conditionality;
- generic status;
- historical status;

- subject or experiencer.

The resulting cTAKES annotations formed the primary candidates for the terminology-centred branch.

4.3.2 Disease-concept selection

A cTAKES mention could be associated with several UMLS concepts. Candidate concepts were therefore scored according to their semantic type, lexical agreement with the extracted phrase, preferred term, terminology source and availability of an official NHS mapping.

*The primary disease-oriented **UMLS semantic types** retained by the implementation were:*

- T019 — Congenital Abnormality;
- T020 — Acquired Abnormality;
- T037 — Injury or Poisoning;
- T047 — Disease or Syndrome;
- T048 — Mental or Behavioural Dysfunction;
- T191 — Neoplastic Process.

These semantic types were used to retain candidate disease and pathological-condition concepts for further processing; they did not guarantee automatic inclusion in the final output. In particular, T037 concepts were retained only when the extracted injury represented an eligible confirmed patient condition under the project's morbidity definition.

Concepts corresponding only to findings, anatomical structures, procedures or medications were excluded. Generic expressions such as disease, problem, mass or history were also rejected when they did not identify a sufficiently specific morbidity.

4.3.3 Assertion and context processing

The Java pipeline interpreted the cTAKES assertion attributes together with additional section- and scope-aware safeguards.

A mention was excluded when it was:

- assigned negative polarity;
- uncertain or conditional;
- hypothetical or contained in patient instructions;
- attributed to a non-patient subject;
- located in a family-history section;
- generic rather than diagnostic;
- located only in an excluded section such as medications, allergies or laboratory data.

Historical patient diseases were retained, as required by the morbidity definition. Additional bidirectional context patterns handled constructions in which negation or uncertainty appeared either before or after the disease expression, such as no evidence of pneumonia and pneumonia is not present.

Consequently, cTAKES did process clinical context, but it did so through assertion attributes and explicit rules rather than through the contextual representation used by a transformer.

4.3.4 Terminology and abbreviation supplements

The final cTAKES-centred pipeline was not limited to the unmodified cTAKES output. A supplementary abbreviation pass recognised common clinical forms such as *HTN*, *COPD*, *CKD*, *DM2* and *NSTEMI*. Context-sensitive rules were applied to ambiguous abbreviations before an expansion was accepted.

A further terminology pass used exact disease expressions from UMLS 2026AA when cTAKES had produced no overlapping disease annotation. This was intended to improve coverage for valid clinical expressions missing from the local cTAKES lookup configuration. The supplement used terminology-based disease expressions rather than a table created from the final evaluation records.

A limited set of spelling normalisations was also applied before concept matching. These additions increased terminology coverage but mean that the final results should be attributed to the complete cTAKES-centred pipeline, not to standard cTAKES alone.

4.3.5 cTAKES deduplication

Deduplication was performed in two stages. First, exact normalised textual repetitions were consolidated within each note. If the same phrase was produced by more than one extraction component, priority was given to the primary cTAKES annotation, followed by the abbreviation supplement and then the additional terminology pass.

After terminology mapping, rows were grouped using SNOMED CT identifiers, UMLS CUIs and compatible normalised concepts. Compatibility guards prevented clinically distinct conditions from being merged.

These guards preserved distinctions including:

- type 1 versus type 2 diabetes;
- acute versus chronic disease;
- left- versus right-sided conditions;
- STEMI versus NSTEMI;
- traumatic versus non-traumatic disease;
- subdural versus epidural conditions.

The representative row was selected using mapping provenance, availability of terminology identifiers, assertion status and extraction source. Alternative surface expressions and the number of merged occurrences were retained for auditing.

4.4 Concept normalisation and SNOMED CT UK linking

Concept normalisation converted different textual expressions representing the same clinical condition into a comparable form. Basic text normalisation standardised case, whitespace and punctuation. Roman numerals were converted only when they occurred in explicit constructions such as disease type, stage, grade or class.

The transformer-centred pipeline attempted to link extracted expressions to active SNOMED CT UK descriptions. Exact normalised SNOMED synonyms and UMLS aliases linked to active UK concepts were preferred. If no exact match was available, a conservative lexical comparison was applied. A lexical match was accepted only when it achieved a sufficiently high score, differed clearly from alternative candidates and preserved clinically important modifiers such as type, stage, anatomical site, laterality and acuity.

The cTAKES-centred pipeline began with its UMLS CUI and preferred term. Where possible, these were linked to a corresponding SNOMED CT UK concept. Additional lexical matching was used when the cTAKES CUI did not provide a unique UK SNOMED CT identifier.

Uncertain concept matches were retained for review rather than silently promoted to confirmed concepts. This was important because an incorrect terminology link could subsequently generate an incorrect ICD-10 suggestion.

4.5 Suggested NHS ICD-10 coding

Eligible concepts were mapped using the official SNOMED CT UK to ICD-10 complex-map reference set and the NHS ICD-10 5th Edition codes and titles [2], [3], [15]. The mapping process did not use ICD-10-CM codes or MIMIC diagnosis-code fields as substitutes for NHS ICD-10.

The pipeline evaluated the mapping evidence associated with each candidate SNOMED CT concept. When the evidence supported one reliable three-character NHS ICD-10 family, that family was recorded as the primary suggestion. Conditional mappings, multiple map blocks and mappings requiring clinical details unavailable in the extracted text were marked for review.

When a unique family could not be determined safely, the pipeline retained a ranked list of up to five candidate ICD-10 families. Candidate families were ranked using:

- support across the linked SNOMED CT concepts;
- lexical similarity to the official ICD-10 family title;
- the priority, block and group ordering provided by the official mapping resource.

Candidate lists were provided for manual review and were not presented as definitive coding decisions. The output should therefore be interpreted as coding support rather than automatic replacement of a qualified clinical coder.

4.6 Structured output and audit information

Both pipelines were converted into a common logical output. Table 4.2 presents the principal fields used in the comparison.

Table 4.2 Common structured output fields

Field	Description
note_id	Discharge-summary identifier
subject_id	Patient identifier
hadm_id	Hospital-admission identifier
morbidity	Extracted textual expression
normalised_morbidity	Standardised expression
clinical_status	Current, historical, uncertain, negated or excluded status
UMLS_CUI	UMLS concept identifier, where available
SNOMED_CT_ID	SNOMED CT UK identifier, where available
ICD_10_suggested	Suggested ICD-10 family or ranked candidates
source_system	Bio_ClinicalBERT-centred or cTAKES-centred pipeline
needs_review	Indicator of unresolved or ambiguous mapping

The Bio_ClinicalBERT workbook separated confirmed diseases, confirmed non-disease conditions, all candidate mentions, excluded mentions and processing statistics. The cTAKES workbook similarly retained accepted morbidity concepts, excluded assertion cases, terminology provenance, mapping candidates and pipeline statistics.

This audit information made it possible to trace a final suggestion back to its textual evidence, contextual decision, terminology link and mapping source. The common output format was then used to construct the manual evaluation workbook described in [Chapter 3](#).

5 Results

This chapter reports the development-stage assessment and final held-out performance of the two implemented pipelines. In the figures, the label EHRKit refers to the complete Bio_ClinicalBERT-centred pipeline rather than to the neural model alone.

5.1 Initial Bio_ClinicalBERT assessment

The initial model-development process selected epoch 7 and an entity-confidence threshold of 0.89. Table 5.1 presents the saved calibration and internal-validation results at the unique-concept level.

Table 5.1 Initial Bio_ClinicalBERT development-stage results

Stage	TP	FP	FN	Precision	Recall	F1
Calibration at threshold 0.89	78	33	46	70.27%	62.9%	66.38%
Initial internal validation	130	81	66	61.61%	66.33%	63.88%

The initial internal-validation F1 score of 63.88% did not meet the pre-specified 80% internal quality gate. The optional refit across all 100 annotated notes was therefore not performed at that stage. Instead, the observed errors motivated the iterative refinement procedure described in Sections 3.5 and 4.2, 4.3, 4.4, 4.5.

The development-stage results should not be compared directly with the final held-out results as if they represented performance before and after a single controlled intervention. They were calculated on different records and reflected different stages of the processing workflow. They nevertheless document why further contextual, terminology, deduplication and coding refinements were necessary.

The cTAKES-centred pipeline was also refined using the separate development records. Its final performance should therefore be compared with that of the Bio_ClinicalBERT-centred pipeline only on the shared held-out records.

5.2 Evaluation dataset

Following pipeline development and refinement, both configurations were frozen. A preliminary paired evaluation was first conducted on 20 previously unseen, patient-disjoint discharge summaries containing 274 Gold-standard morbidity concepts. The evaluation was subsequently expanded to 200 previously unseen discharge summaries, producing a substantially larger Gold standard of 2,103 morbidity concepts. No model training, threshold selection, terminology refinement or rule modification was performed using either evaluation set, and the same morbidity definition, concept-matching rules and scoring procedure were retained throughout.

Table 5.2 Evaluation datasets

Evaluation	Records	Gold morbidity concepts	Role
Preliminary evaluation	20	274	Initial paired assessment
Expanded evaluation	200	2,103	Larger-scale robustness assessment

In the preliminary 20-record evaluation, the **Bio_ClinicalBERT**-centred pipeline produced **213** unique morbidity predictions, comprising **194** true positives and **19** false positives, while **80** Gold-standard concepts were missed. The **cTAKES**-centred pipeline produced **266** predictions, comprising **216** true positives and **50** false positives, with **58** false negatives.

In the expanded 200-record evaluation, the same operating pattern remained evident. The **Bio_ClinicalBERT**-centred pipeline produced **1,530** unique morbidity predictions, of which **1,378** were true positives and **152** were false positives; **725** Gold-standard morbidities were missed. The **cTAKES**-centred pipeline produced **2,220** predictions, including **1,590** true positives and **630** false positives, while **513** Gold-standard morbidities were missed.

The larger evaluation therefore preserved the distinction observed in the preliminary experiment. *Bio_ClinicalBERT continued to return a smaller and more conservative set of morbidity predictions, whereas cTAKES identified a larger number of Gold-standard conditions but also generated substantially more false-positive concepts.* The corresponding precision, recall and F1 results are presented in Section 5.3.

5.3 Morbidity-extraction results

[Table 5.3](#) reports precision, recall and F1 for both the preliminary 20-record evaluation and the expanded 200-record evaluation, while [Figure 5.1](#) provides a visual comparison of the two systems across both evaluation sizes.

Table 5.3 Morbidity-extraction performance across the two evaluations

System	Evaluation	Precision	Recall	F1
Bio_ClinicalBERT	20 records	91.08%	70.8%	79.67%
Bio_ClinicalBERT	200 records	90.07%	65.53%	75.86%
cTAKES	20 records	81.2%	78.83%	80%
cTAKES	200 records	71.62%	75.61%	73.56%

The preliminary evaluation already indicated contrasting operating profiles. Bio_ClinicalBERT achieved substantially higher precision (91.08%) but lower recall (70.80%), whereas cTAKES achieved lower precision (81.20%) but higher recall (78.83%). Their F1 scores were almost identical at this stage (79.67% and 80.00%, respectively), indicating that the two pipelines reached a similar overall balance through different precision–recall trade-offs.

The expanded 200-record evaluation preserved the same distinction between the systems. Bio_ClinicalBERT achieved a precision of 90.07%, recall of 65.53% and F1 of 75.86%, while cTAKES achieved 71.62% precision, 75.61% recall and 73.56% F1. On the larger evaluation set, Bio_ClinicalBERT therefore had 18.45 percentage points higher precision, whereas cTAKES had 10.08 percentage points higher recall. Bio_ClinicalBERT achieved the higher F1 score by 2.30 percentage points.

Performance estimates were lower in the expanded evaluation, but the relative behaviour of the pipelines remained consistent. Bio_ClinicalBERT precision changed only slightly, from 91.08% to 90.07% (–1.01 percentage points), while its recall decreased from 70.80% to 65.53% (–5.27 percentage points). cTAKES showed a larger reduction in precision, from 81.20% to 71.62% (–9.58 percentage points), while recall decreased more moderately from 78.83% to 75.61% (–3.22 percentage points). F1 decreased by 3.81 percentage points for Bio_ClinicalBERT and by 6.44 percentage points for cTAKES.

Overall, the expanded evaluation strengthened the interpretation established by the preliminary experiment: **Bio_ClinicalBERT remained the more selective and precision-oriented pipeline, whereas cTAKES continued to retrieve a broader proportion of the Gold-standard morbidities**. The larger sample therefore provided a more robust estimate of absolute performance while retaining the same underlying precision–recall trade-off.

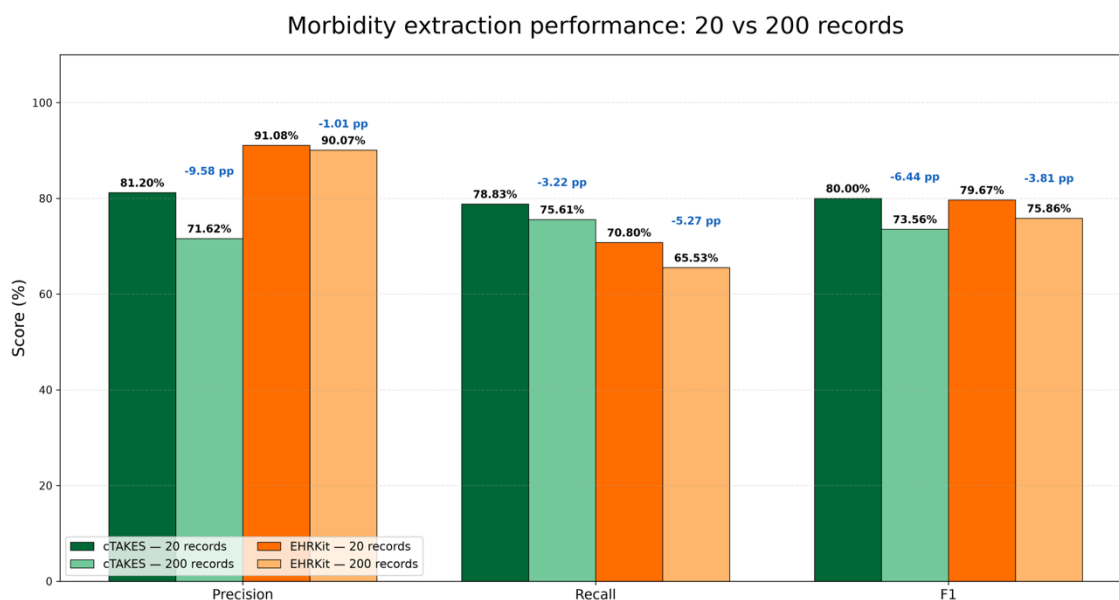


Figure 5.1 Morbidity-extraction performance in the preliminary 20-record and expanded 200-record evaluations

The expanded evaluation produced more conservative absolute performance estimates but preserved the system behaviour observed in the preliminary experiment. *Bio_ClinicalBERT* remained strongly precision-oriented: its precision changed only slightly, from 91.08% to 90.07%, while recall decreased from 70.80% to 65.53%. *cTAKES* continued to achieve higher recall, although its precision decreased more substantially on the larger sample. The expanded evaluation therefore confirmed that the distinction between the pipelines was not an artefact of the initial 20-record sample: *Bio_ClinicalBERT* remained more selective, whereas *cTAKES* continued to retrieve a broader proportion of the Gold-standard morbidities.

5.4 ICD-10 coding results

Table 5.3 presents the underlying coding counts and calculated metrics, while Figure 5.2 visually compares coding coverage, correct-code coverage and coding accuracy.

Table 5.4 Suggested NHS ICD-10 coding performance

System	Correctly extracted morbidities	Morbidities receiving a code	Correct codes	Coding coverage	Correct-code coverage	Coding accuracy
Bio_Clinical-BERT pipeline	194	175	165	90.21%	85.05%	94.29%
cTAKES pipeline	216	216	207	100%	95.83%	95.83%

The **cTAKES**-centred pipeline provided an ICD-10 suggestion for all 216 correctly extracted morbidity concepts, producing coding coverage of 100%. Of these suggestions, 207 were correct, resulting in both correct-code coverage and coding accuracy of 95.83%.

The **Bio_ClinicalBERT**-centred pipeline assigned a code to 175 of its 194 true-positive morbidities, producing coverage of 90.21%. A total of 165 assigned codes were correct. Therefore, its correct-code coverage was 85.05%, while its coding accuracy among assigned codes was 94.29%.

The distinction between these measures is important. Bio_ClinicalBERT's high coding accuracy indicates that its suggestions were usually correct when it produced one. Its lower correct-code coverage was caused primarily by 19 correctly extracted morbidities that did not receive a code. A further 10 assigned suggestions were incorrect. In comparison, cTAKES assigned a code to every true-positive morbidity, although nine of its suggestions were incorrect.

The difference in coding accuracy between the systems was only 1.54 percentage points. The larger differences occurred in coding coverage and correct-code coverage. This indicates that the main coding advantage of the cTAKES-centred pipeline was its ability to generate a suggestion more consistently, rather than a substantial difference in the correctness of assigned suggestions.

A coding coverage of 100% does not mean that cTAKES found every gold-standard morbidity or coded every prediction correctly. It means only that every morbidity counted as a cTAKES true positive received an ICD-10 suggestion.

Figure 5.2. Suggested NHS ICD-10 coding performance of the two pipelines on the final 20 held-out discharge summaries. Coding metrics were calculated for correctly extracted morbidity concepts.

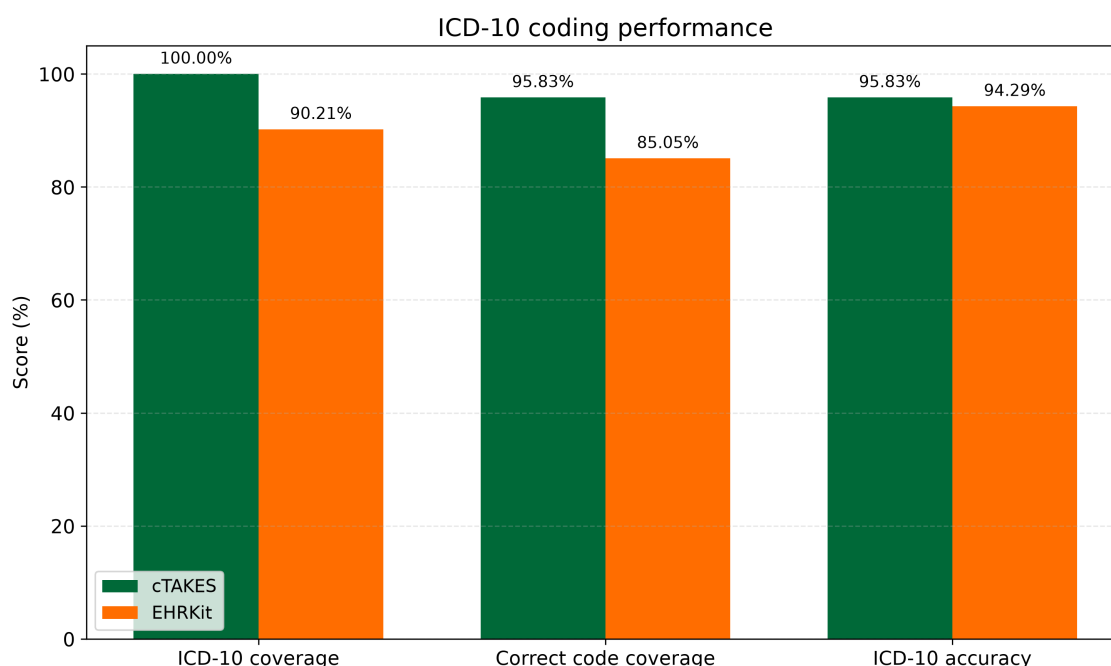


Figure 5.2 ICD-10 coding coverage, correct-code coverage and coding accuracy on the final 20 records

5.5 Error analysis

5.5.1 Bio_ClinicalBERT-centred pipeline

The principal quantitative limitation of the Bio_ClinicalBERT-centred pipeline was its number of false negatives. It **missed 80 of the 274 gold-standard morbidity concepts**. Development and output review indicated several possible sources:

- uncommon diseases insufficiently represented in the annotated training data;

- abbreviations and spelling variants not learned or normalised successfully;
- predicted spans falling below the 0.89 confidence threshold;
- incomplete or incorrect span boundaries;
- information divided across long-document windows;
- eligible conditions removed by conservative contextual or semantic filtering.

The high confidence threshold and strict post-processing rules reduced false positives but could also reject valid low-confidence conditions. This helps explain the combination of high precision and lower recall.

The pipeline **produced only 19 false positives**. Remaining errors included contextually ambiguous mentions, terms classified as diseases despite representing findings, and cases in which a predicted span did not correspond to the manually defined morbidity concept.

5.5.2 cTAKES-centred pipeline

The cTAKES-centred pipeline **missed 58 gold-standard morbidities** but **produced 50 false positives**. Its greater recall was supported by its broad terminology resources, abbreviation supplement and exact UMLS disease-term pass.

The same broad coverage also introduced several potential error sources:

- dictionary matches that were medically valid concepts but not eligible patient morbidities;
- ambiguous abbreviations assigned an incorrect disease meaning;
- broad UMLS disease categories that did not match the intended gold-standard concept;
- assertion modifiers extending beyond their intended scope;
- mentions in historical, instructional or non-diagnostic contexts being interpreted incorrectly;
- several candidate concepts being available for the same surface expression.

The custom assertion and semantic safeguards reduced these problems but did not eliminate them. This explains why cTAKES achieved higher recall while retaining lower precision than the transformer-centred pipeline.

5.5.3 Terminology and ICD-10 mapping errors

Nineteen Bio_ClinicalBERT true positives did not receive an ICD-10 suggestion, whereas **every cTAKES true positive received at least one suggestion**.

Missing or incorrect mappings occurred when:

- an extracted phrase could not be linked confidently to an active SNOMED CT UK concept;
- several SNOMED CT concepts matched the same expression;
- the clinical text lacked information about disease type, stage, cause, anatomical site or complication;
- the official map contained conditional rules or multiple map blocks;

- the terminology concept and ICD-10 family represented different levels of clinical detail.

The ranked candidate lists preserved possible mappings for review, but an available candidate was not necessarily considered a correct code during evaluation.

Because the evaluation used three-character ICD-10 families, the reported coding accuracy does not establish accuracy for complete four- or five-character coding. More specific coding would require additional clinical detail and validation by a qualified clinical coder.

5.6 Results summary

Neither pipeline performed best across all measures. The **Bio_ClinicalBERT-centred pipeline achieved higher precision** and produced fewer false-positive morbidities. The **cTAKES-centred pipeline achieved higher recall, coding coverage and correct-code coverage**. Their observed **F1 scores and coding accuracies were similar**.

These results apply to the complete customised pipelines evaluated in this project. They should not be interpreted as a direct benchmark of unmodified Bio_ClinicalBERT against standard Apache cTAKES. Their implications, relationship to previous research, limitations and potential use in a hybrid system are considered in Chapter 6.

6 Evaluation and Conclusions

This chapter evaluates the extent to which the project answered its research questions and achieved its objectives. It also considers the contribution of the findings in relation to existing work, identifies the principal limitations and proposes directions for further development.

6.1 Summary

This project developed and evaluated two end-to-end pipelines for extracting patient morbidities from complete MIMIC-IV discharge summaries and assigning suggested NHS ICD-10 5th Edition families. The first pipeline combined a fine-tuned Bio_ClinicalBERT sequence tagger with contextual, semantic and terminology-based post-processing. The second combined Apache cTAKES with UMLS concept extraction, assertion processing, terminology supplementation and UK-specific mapping.

A patient-disjoint collection of discharge summaries was used across model development, pipeline refinement and final evaluation. The Bio_ClinicalBERT development corpus contained 100 manually annotated notes, while 60 additional records supported iterative refinement of both pipelines. The final paired comparison used 20 held-out records containing 274 unique gold-standard morbidity concepts.

The Bio_ClinicalBERT-centred pipeline achieved 91.08% precision, 70.80% recall and 79.67% F1. The cTAKES-centred pipeline achieved 81.20% precision, 78.83% recall and 80.00% F1. Bio_ClinicalBERT therefore produced fewer false-positive morbidities, whereas cTAKES identified a larger proportion of the gold-standard conditions. The observed F1 difference was only 0.33 percentage points and does not demonstrate clear overall superiority.

For suggested ICD-10 mapping, Bio_ClinicalBERT achieved 90.21% coding coverage, 85.05% correct-code coverage and 94.29% coding accuracy. cTAKES achieved 100% coding coverage and 95.83% for both correct-code coverage and coding accuracy. Both systems usually produced a correct family when a suggestion was assigned, but cTAKES provided suggestions more consistently.

6.2 Evaluation of the project

6.2.1 Evaluation against research questions

1. ***The first research question** asked how accurately the Bio_ClinicalBERT-centred and cTAKES-centred pipelines could extract unique morbidity concepts from complete MIMIC-IV discharge summaries.*

The results do not identify one system as superior across all extraction measures. Bio_ClinicalBERT achieved higher precision and produced only 19 false positives, compared with 50 for cTAKES. This indicates that the transformer-centred pipeline was more conservative and that a larger proportion of its extracted morbidities were correct.

cTAKES achieved higher recall and identified 216 of the 274 gold-standard morbidity concepts, compared with 194 identified by Bio_ClinicalBERT. It therefore produced a more complete morbidity list, although this increased coverage was accompanied by more false-positive predictions.

The nearly identical F1 scores show that the two systems achieved a similar overall balance between precision and recall through different operating behaviours. As no confidence intervals or statistical significance tests were calculated, the 0.33-percentage-point F1 difference should be treated as descriptive rather than evidence that cTAKES is generally more accurate.

2. *The second research question asked how effectively the extracted morbidities could be normalised and assigned suggested NHS ICD-10 5th Edition families.*

Both pipelines achieved high coding accuracy for assigned suggestions. Bio_ClinicalBERT achieved 94.29% accuracy, while cTAKES achieved 95.83%. This indicates that most code suggestions were correct once an eligible morbidity had been linked successfully.

The greater difference concerned coding coverage. cTAKES assigned a suggestion to every correctly extracted morbidity, whereas Bio_ClinicalBERT assigned suggestions to 175 of its 194 true-positive morbidities. The corresponding correct-code coverage values were 95.83% and 85.05%. **cTAKES was therefore more successful at providing a usable code suggestion consistently.**

However, these results relate to suggested three-character code families rather than definitive clinical coding. Some SNOMED CT concepts produced conditional mappings, multiple possible targets or mappings requiring clinical details unavailable in the extracted phrase.

3. *The third research question examined the complementary strengths, limitations and error patterns of the transformer-based and terminology-driven approaches.*

The principal **strength of the Bio_ClinicalBERT-centred pipeline was precision.** Its learned representation, confidence threshold and contextual post-processing enabled it to reject many inappropriate disease candidates. Its **principal limitation was recall:** uncommon conditions, abbreviations and low-confidence mentions were more likely to be missed.

The principal **strength of the cTAKES-centred pipeline was terminology coverage.** Its UMLS resources, abbreviation handling and supplementary terminology pass allowed it to identify more gold-standard morbidities and provide broader coding coverage. Its principal **limitation was the number of false positives** generated by broad dictionary matches, ambiguous abbreviations and imperfect assertion or semantic filtering.

The research questions were therefore answered through the identification of a clear precision–recall trade-off. **The systems achieved similar F1 scores but provided different practical advantages.**

6.2.2 Evaluation against the project objectives

Table 6.1 evaluates the work against the objectives defined in Section 1.3.

Table 6.1 Evaluation of the project objectives

Objective	Evidence of achievement	Outcome
Prepare MIMIC-IV discharge summaries and create a manually annotated gold standard	Complete records were selected, annotated and separated at patient level; the final gold standard contained 274 unique concepts	Achieved
Develop and fine-tune a Bio_ClinicalBERT morbidity-extraction pipeline within EHRKit	A BIO token-classification model was trained, calibrated and integrated with context, terminology and coding components	Achieved
Configure cTAKES for morbidity extraction from the same type of records	cTAKES 6.0.0 was integrated with UMLS extraction, assertion processing, terminology supplementation and UK mapping	Achieved

Apply context filtering and within-record deduplication	Both pipelines excluded ineligible contexts and consolidated repeated or synonymous concepts	Achieved
Normalise morbidities and map them to suggested NHS ICD-10 families	UMLS and SNOMED CT UK resources were linked to NHS ICD-10 mapping resources	Achieved with limitations
Compare the pipelines using precision, recall and F1	Both frozen pipelines processed the same final records and were assessed using identical scoring rules	Achieved
Evaluate coding coverage and accuracy and analyse system strengths and limitations	Coding coverage, correct-code coverage and coding accuracy were calculated, followed by quantitative and qualitative error analysis	Achieved

The objectives were therefore substantially achieved. The qualification applied to terminology normalisation and coding reflects the fact that some concepts did not receive an ICD-10 coding suggestion and that suggested code families were not validated as definitive codes by qualified clinical coders.

The project followed a controlled and reproducible evaluation design. As described in Chapter 3, patient-disjoint development and final-evaluation data were separated, and the final held-out records were excluded from model selection and pipeline refinement. Both frozen pipelines were evaluated on the same clinical texts against the same gold standard using consistent matching and scoring rules. Saved partition information and auditable system outputs supported reproducibility and traceability, while the initial internal-validation result was reported transparently rather than omitted. These controls strengthen confidence in the final comparison; however, the systems remain research prototypes and would require further validation before clinical use.

6.3 Comparison with Existing Work

The results are consistent with previous research showing that clinical concept-recognition performance depends strongly on the dataset, entity definition, terminology coverage and evaluation method. Lossio-Ventura et al. reported substantial variation among clinical concept-recognition systems across exact and inexact matching, negation, abbreviations, ambiguity and misspellings [8]. *The present study similarly found that no system was optimal across all evaluation measures.*

The original cTAKES work demonstrated the benefits of combining clinical dictionaries, UMLS concept mapping and contextual attributes in a modular clinical NLP architecture [5]. *The current project confirms that this approach can provide broad disease and terminology coverage when applied to complete discharge summaries.* cTAKES achieved the higher recall and complete code-assignment coverage. However, its lower precision demonstrates the continuing risk that broad terminology matching will identify medically valid expressions that do not represent eligible patient morbidities.

The reported cTAKES results must be interpreted as the performance of the complete customised pipeline. Additional assertion safeguards, abbreviation recognition, exact UMLS terminology supplementation, semantic filtering and UK-specific mapping were applied around the standard cTAKES components. *The results should therefore not be interpreted as a benchmark of the unmodified cTAKES distribution.*

Previous research on Bio_ClinicalBERT and related biomedical transformers has shown that **domain-specific pretraining can support clinical entity extraction by providing representations adapted to biomedical and clinical language** [6], [11]. *The higher precision achieved by the transformer-centred pipeline is consistent with the ability of contextual representations to distinguish relevant disease mentions from surrounding text.*

Nevertheless, the *lower recall demonstrates that clinical pretraining alone does not ensure complete morbidity extraction.* Performance remained dependent on the size and coverage of the manually annotated fine-tuning corpus, the selected confidence threshold and the processing of long documents. Furthermore, the final result included medSpaCy ConText, section rules, semantic filtering and terminology linking. The higher precision cannot therefore be attributed exclusively to Bio_ClinicalBERT.

MedCAT represents an existing example of combining statistical learning with terminology-based concept linking [9]. Its design supports the broader argument that contextual representations and medical terminologies can provide complementary information. *The present study contributes additional task-specific evidence for this argument by demonstrating different precision and recall profiles under the same morbidity definition and evaluation conditions.*

Research on automatic ICD coding frequently treats code prediction as a multi-label document-classification problem. CAML, for example, predicts a collection of ICD codes from a complete clinical document using label-specific attention [13]. BERT-based coding research has similarly considered the challenges of long documents, large label spaces and rare codes [12]. More recent work has also investigated generative models for discharge-summary coding [14]. *The present project used a different design.* It first extracted individual morbidity concepts, determined their clinical context and then produced a code suggestion associated with each retained morbidity. This provides a more explicit relationship between the clinical evidence and the suggested code. It also permits separate measurement of extraction performance, coding coverage and coding accuracy.

The mapping results confirm the position of NHS and SNOMED guidelines that mapping between SNOMED CT and ICD-10 codes is not always a simple one-to-one operation [2], [15]. *Some mappings depended on information about disease type, stage, cause, anatomical site or complications.* The use of ranked candidates and review indicators was therefore more appropriate than presenting every mapping as definitive.

Direct numerical comparison with F1 scores or coding accuracy reported by other studies would not be appropriate because the datasets, entity definitions, annotation policies, code systems and matching rules differ. *The main contribution of this project is the controlled head-to-head comparison of two substantially different end-to-end approaches on the same complete records, using a common concept-level gold standard and the same evaluation procedure.*

6.4 Limitations and future work

6.4.1 Limitations

A few limitations should be considered when interpreting the results.

1. The final evaluation set contained only 20 discharge summaries and 274 unique morbidity concepts. Although the records were patient-disjoint, the **sample was too small** to establish precise population-level performance estimates. Consequently, the small observed F1 difference cannot establish general superiority.
2. The clinical records were obtained from MIMIC-IV, which represents de-identified data from a United States healthcare organisation. **Documentation practices, abbreviations and terminology may differ across hospitals and countries.**

3. The two evaluated systems were customised end-to-end pipelines. Bio_ClinicalBERT was combined with contextual and terminology-based post-processing, while cTAKES was supplemented with additional abbreviation, assertion and terminology components. **The results cannot be generalised directly to unmodified Bio_ClinicalBERT, EHRKit or cTAKES installations.**

4. The **Bio_ClinicalBERT model was developed using a relatively small annotated corpus.** Rare diseases, uncommon abbreviations and unusual language patterns may have been insufficiently represented. Its processing of long notes through overlapping windows could also miss relationships between information appearing in distant parts of a discharge summary.

5. The coding evaluation was conducted at the level of **three-character ICD-10 families.** It did not establish accuracy for complete four- or five-character coding.

6. Finally, the systems were evaluated as **research tools rather than within a live clinical workflow.** Runtime, integration with an operational EHR, user interaction, coder workload and the consequences of false-positive or false-negative conditions were not evaluated.

6.4.2 Future work

- Future work should evaluate both pipelines on a substantially larger collection of patient-disjoint records.
- Data should ideally be obtained from multiple healthcare organisations and should include clinical text produced within the NHS.
- The gold standard should be created independently by at least two annotators with appropriate clinical or coding expertise.
- Bio_ClinicalBERT could be fine-tuned on a larger and more diverse annotated dataset.
- A further improvement would be to introduce a dedicated coding model as an additional layer after concept normalisation. This model could rank only those ICD-10 families supported by the official NHS terminology and mapping resources. Its inputs could include the extracted morbidity, surrounding clinical context, linked SNOMED CT concept and mapping attributes. The model should retain the ability to abstain or return a short candidate list when the available evidence is insufficient for a reliable primary suggestion.

6.4.3 Hybrid morbidity-extraction pipeline

The complementary precision and recall profiles suggest that a hybrid pipeline is a promising direction. However, it cannot be assumed that combining the systems will automatically produce a higher F1 score. *A hybrid design must be implemented and evaluated on new held-out records.*

One possible architecture would use cTAKES as a high-coverage candidate generator. Its disease mentions, UMLS CUIs, assertion attributes and terminology variants would be transferred to the Python pipeline using JSON, CSV or a service interface. Bio_ClinicalBERT would then examine each candidate together with its surrounding clinical context and estimate whether it represented an eligible patient morbidity.

The transformer could reject inappropriate dictionary matches caused by negation, uncertainty, family history, ambiguous abbreviations or non-diagnostic sections. Independently predicted high-confidence Bio_ClinicalBERT spans could then be added when they did not overlap a cTAKES candidate. The combined concepts would be normalised and deduplicated before coding.

Figure 6.1 presents the proposed hybrid process.

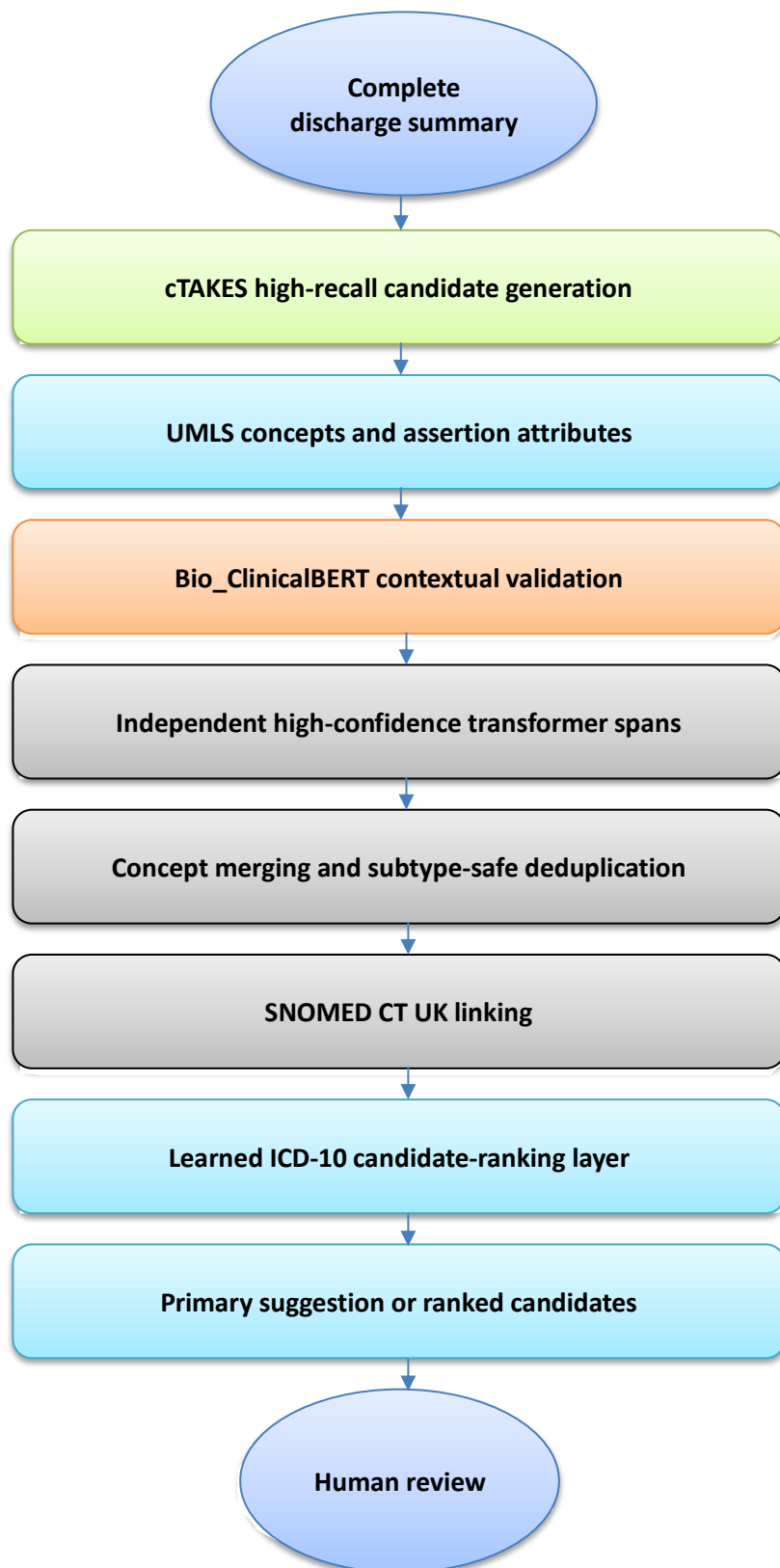


Figure 6.1 Proposed hybrid morbidity-extraction and coding architecture

The use of Python for Bio_ClinicalBERT and Java for cTAKES does not prevent this design. Both systems can exchange mention-level data through a language-independent format containing text offsets, candidate concepts, confidence scores and contextual attributes.

The hybrid system should be evaluated against each individual pipeline using the same patient-disjoint gold standard. In addition to precision, recall and F1, the evaluation should measure how frequently one system corrects an error made by the other and whether candidate validation introduces new false negatives.

6.5 Conclusion

This dissertation compares a transformer-centred Bio_ClinicalBERT pipeline with a terminology-centred cTAKES pipeline for extracting unique patient morbidities from complete MIMIC-IV discharge summaries and assigning suggested NHS ICD-10 families.

Bio_ClinicalBERT achieved higher precision and produced fewer false-positive conditions. cTAKES achieved higher recall and more complete coding coverage. Their nearly identical observed F1 scores provide no evidence that either approach was uniformly superior under the evaluated conditions. Instead, the systems provided complementary capabilities: contextual selectivity from the transformer-centred workflow and broad terminology coverage from the cTAKES-centred workflow.

On the evaluated records, the project demonstrates that free-text discharge summaries can be converted into structured, concept-level morbidity information with high precision and useful coding accuracy. It also shows that extraction and coding should be evaluated separately, because identifying the correct morbidity does not guarantee that an appropriate code will be available.

The work provides a reproducible basis for further development of a hybrid system combining terminology-based candidate generation, transformer-based contextual validation and learned ICD-10 candidate ranking. Such a system could potentially provide more complete and reliable morbidity summaries, but its benefit would need to be confirmed through larger-scale evaluation using independently annotated clinical notes.

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